

[DC] Leveraging AR Cues towards New Navigation Assistant Paradigm

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ABSTRACT

Extensive research has shown that the knowledge required to navigate an unfamiliar environment has been greatly reduced as many of the planning and decision-making tasks can be supplanted by the use of automated navigation systems. The progress in augmented reality (AR), particularly AR head-mounted displays (HMDs) foreshadows the prevalence of such devices as computational platforms of the future. AR displays open a new design space on navigational aids for solving this problem by superimposing virtual imagery over the environment. This dissertation abstract proposes a research agenda that investigates how to effectively leverage AR cues to help both navigation efficiency and spatial learning in walking scenarios.

1 INTRODUCTION

People utilize navigation systems in their daily life to support wayfinding through the environment. Current navigational aid design focuses on usability issues such as interface design, interaction techniques, and problems of divided attention. Although usability is vital in navigation system design, in line with *cognitive engineering*, designers and developers should also take into account how design elements interact with cognitive processes, with an emphasis on spatial information processing and individual differences in spatial ability. However, research has established that navigation system utilization, especially turn-by-turn instructions, is associated with degraded spatial learning. This dissertation proposal focuses on the design of navigation assistance systems emphasizing spatial learning.

Augmented reality (AR) is a promising technology for navigation systems because it can integrate virtual content into the real world, reducing the need for divided attention, or it can provide graphical overlays of information when that is more useful. The promise of AR is that the technology will evolve to become one that users can wear throughout the day. Currently, we use head-mounted displays (HMDs) to test AR designs for the future. HMD-based AR systems for indoor and outdoor use are increasing. Yet, little research has been conducted on designing pedestrian wayfinding and navigation assistance in AR HMDs. In this research agenda, I develop and evaluate a novel navigation assistance paradigm for AR HMDs that considers both usability and spatial learning.

2 PROJECT FRAMEWORK

Simulated AR systems in virtual reality (VR) have been successfully employed as a valid alternative to a real AR system in areas such as military training, general search tasks, and usability evaluation. Therefore, in the first two research questions (RQ) in my dissertation, I plan to investigate the AR navigation interface design in VR. This confronts participants with unknown environments in a tightly controlled study setup for fairly assessing spatial knowledge acquisition. In RQ3, I will apply what I find in RQ1 and RQ2 in VR to AR HMDs to evaluate the usability of the design for AR HMDs (the intention is to use an optical see-through device, the Microsoft

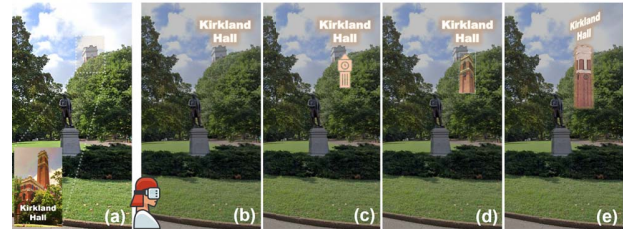


Figure 1: Illustrations of a Virtual Landmark (VL) with different visualization complexity in AR HMDs. (a) A partially occluded landmark; VL is facing the user and represented in (b) text, (c) text + 2D geometry, (d) text + image; VL is aligned with the landmark facade and represented in (e) text + 3D models.

HoloLens 2) and get further insights. This also contributes to evaluating whether and how VR can be used as an effective testbed to rapidly prototype and assess designs for AR.

RQ1. Exploring the interaction between design view and types of information in pedestrian AR navigation systems

Following research in AR interfaces for automobile, the design space for AR can be categorized into *screen-fixed* and *world-fixed* [5] views. Screen-fixed designs present visuals that are in a constant position with respect to the user's head, whereas world-fixed designs present visuals that have position relative to the global coordinate system. Traditionally, the AR community has held the view that world-fixed designs are better in terms of spatial knowledge acquisition but recently some work has challenged this view [4]. Besides, the type of information to display in navigation is also a significant factor to consider. The two most important pieces of information that navigation guidance systems should provide are *directional* and *landmark* information [8]. Directional information plays a significant part in navigation completion; landmarks are the most predominantly used navigation cue in pedestrian navigation instructions. Direction information in AR navigation systems typically consists of virtual cues that indicate a direction to go or an orientation; landmark information consists of virtual cues that highlight a recognizable environmental feature that could serve as a landmark or a virtual object that could serve as a landmark. Previous work has shown that including landmarks in navigation instructions as essential environmental cues have a positive effect on navigation performance and spatial learning [2, 6].

The open question for AR pedestrian navigation system design is which type of information is best suited to what type of views. For instance, does directional information work best in a screen-fixed or world-fixed system? Likewise, does landmark information work best in a screen-fixed or world-fixed system? And, finally, do these two types of information interfere with one another?

To answer these questions, I propose a 2×2 mixed design experiment varying the factors of view (world-fixed and screen-fixed) and information type (directional and landmark). The experiments will be conducted in VR by modeling photo-realism environments and simulating the limited field of view (FOV) of AR display. I will collect the navigation performance, survey knowledge acquisition by the pointing task, and cognitive load by the detection response task

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to assess the effects of the design view and navigation information. I will follow this with an experiment in which a system combining the best performing landmark and directional systems is compared against individual systems to see how they might interact with one another. The design of this experiment will be informed by the results of the first experiment.

My hypotheses are: (1) people experience best navigation performance gains using world-fixed design; (2) landmark AR cues compared with only directional guidance is more beneficial to spatial learning; and (3) directional and landmark AR cues in world-fixed view would be the optimal interface.

RQ2. Creating effective virtual landmark cues to improve spatial learning while navigation If the world-fixed AR annotation with added landmark cues benefits wayfinding and spatial learning, my next research question is how to effectively design the world-registered landmark AR cues?

Crede et al. [3] suggested that memory is more accurate for *global* (i.e., distal) than for *local* (i.e., proximal) landmark configurations especially for participants with high working memory capacity. Li et al. [7] demonstrated that visualization of distant landmarks on a mobile map display, in comparison with the visualization without distant landmarks, contributes positively to persons with low sense of direction being spatially oriented. As opposed to 2D map-based applications on smartphones, AR can incorporate the wayfinding guidance into the physical environment by situated visualization. It enables developers to display more information without overcrowding compared to the screen space design by offloading information to the surroundings. Singh et al. [9] presented a new Virtual Global Landmark (VGL) technique for an AR-based navigation system that allows seeing a global landmark even if its surroundings hinder it. The VGL compared to traditional means of navigation (i.e., directional arrow) could help users learn more about the environment without any significant mental workload increase.

While their work only included one global landmark with the visualization of a simple 2D drawing of the contour and compared the efficacy of VGL to the directional arrow, I will explore the benefit of combining the directional information and multiple virtual landmark cues in navigation tasks, and the visualization complexity of the virtual landmarks, and how they affect spatial learning and situational awareness of people with different spatial abilities and working memory capacities.

Virtual landmarks could be manipulated in terms of the location complexity (i.e., registration based on only position vs. position and orientation) and content complexity (e.g., text, 2D geometry/image, simple or complex 3D objects) [10] to determine an effective visualization (Figure 1). In addition to creating virtual landmark counterparts of global landmarks, I will also examine the effects of visualizing local landmarks as virtual global landmarks on spatial learning compared to only annotating these local landmarks if they are in the vicinity of the user. Considering the restricted FOV of AR displays, body/screen-fixed auxiliary cues will be used to guide periphery information access.

I predict that (1) visualizing local landmarks as virtual global landmarks will enhance spatial knowledge acquisition particularly for individuals with high working memory capacity and (2) visualization modes with lower complexity would lead to less cognitive workload and maintain spatial learning efficiency. The results will provide design guidelines for future learning-aware navigation systems to optimize AR cue display by considering individual abilities.

RQ3. Prototyping proof-of-concept interfaces in AR HMDs to understand the usability in the wild I will prototype an AR navigation interface on HoloLens 2 considering the insights from RQ1 and RQ2. I am currently working on transmitting GNSS and IMU data into HoloLens and implementing a Kalman Filter and World Locking Tools [1] to support more accurate large-scale outdoor registration. The HoloLens 2 will have difficulty with

the visibility of virtual objects in bright sunlight, and we intend to use a shade over the visor for that.

Participants will be asked to follow a planned route outside and guided by explicit directions and AR landmark cues designed in the RQ1 and RQ2 during route following. After reaching the target, participants will be required to independently navigate back to the starting location. In addition to assessing the objective metrics as in previous experiments, I will collect more subjective feedback (e.g., usability and trust in the system) by semi-structured interviews. I expect that (1) people will be able to effectively follow the intended routes guided by world-registered AR cues without the help of a map; (2) landmarks along the routes in the physical world as well as virtual landmarks displayed in AR can be learned; (3) individuals with high working memory capacity will learn landmarks better and perform better overall.

3 CONCLUSION

Navigation is one of the most common tasks that people perform every day. The design guidelines gleaned from my work that consider the trade-off between navigational efficiency benefits and spatial learning cost may inform not only the design and development of future navigational aids in the real world but also in virtual environments when people are training, gaming or entertaining. I hope my work would call for and contribute to learning-aware AR design at the outset before AR become a widespread technology. I would like to outline some of the questions that could be discussed during the consortium that will better shape the work: (1) What should I consider for deploying an efficient simulated AR in immersive VR to inform software design on future AR platforms that have more advanced specifications? (2) What is an appropriate locomotion method in VR in the context of navigation-related tasks? (3) What other world-space design parameters could be included to assist navigation performance and improve acquisition of spatial knowledge that I have not thought about?

REFERENCES

- [1] World Locking Tools for Unity. <https://microsoft.github.io/MixedReality-WorldLockingTools-Unity>. Accessed: 2021-12-10.
- [2] A. Bolton, G. Burnett, and D. R. Large. An investigation of augmented reality presentations of landmark-based navigation using a head-up display. In *Proc. of the 7th Int. Conf. on Automotive User Interfaces and Interactive Vehicular Applications*, pp. 56–63, 2015.
- [3] S. Credé, T. Thrash, C. Hölscher, and S. I. Fabrikant. The advantage of globally visible landmarks for spatial learning. *Journal of Environmental Psychology*, 67:101369, 2020.
- [4] N. De Oliveira Faria. *Effects of Augmented Reality Head-up Display Graphics' Perceptual Form on Driver Spatial Knowledge Acquisition*. PhD thesis, Virginia Tech, 2019.
- [5] J. L. Gabbard, G. M. Fitch, and H. Kim. Behind the glass: Driver challenges and opportunities for ar automotive applications. *Proceedings of the IEEE*, 102(2):124–136, 2014.
- [6] K. Gramann, P. Hoepner, and K. Karrer-Gauss. Modified navigation instructions for spatial navigation assistance systems lead to incidental spatial learning. *Frontiers in psychology*, 8:193, 2017.
- [7] R. Li, A. Korda, M. Radtke, and A. Schwering. Visualising distant off-screen landmarks on mobile devices to support spatial orientation. *Journal of Location Based Services*, 8(3):166–178, 2014.
- [8] A. J. May, T. Ross, S. H. Bayer, and M. J. Tarkiainen. Pedestrian navigation aids: information requirements and design implications. *Personal and Ubiquitous Computing*, 7(6):331–338, 2003.
- [9] A. K. Singh, J. Liu, C. A. Tirado Cortes, and C.-T. Lin. Virtual global landmark: An augmented reality technique to improve spatial navigation learning. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, pp. 1–6, 2021.
- [10] J. Wither, S. DiVerdi, and T. Höllerer. Annotation in outdoor augmented reality. *Computers & Graphics*, 33(6):679–689, 2009.